MANUAL VERIFICATION

1) 8 bit Addition: 8BH + 65H

8BH = 10001011
65H = 01100101

```
  1000 1011
(+) 0110 0101
-------------
  1111 0000
-------------
```

Result = F0H

CY=0 Z=0
AC=1 S=1
P=1 OV=0

Flag byte (OV, S, Z, AC, P, CY) = 1001 0100 = 94H.

2) 8-Bit Substraction: 85H - 36H

85H = 1000 0101
36H = 0011 0110

```
  1000 0101
(-) 0011 0110
-------------
  0100 1111
-------------
```

Result = 4F1H

CY=0 Z=0
AC=1 S=0
P=0 OV=1

Flag byte (OV, S, Z, AC, P, CY) = 0101 0000 = 50H.

3) 16 bit Addition : 2345H + 4567H

$$\begin{array}{r} 2345H \\ (+)\ 4567H \\ \hline 68ACH \end{array}$$

5+7 = C

4+6 = A

3+5 = 8

2+4 = 6

Result: 68ACH, no carry beyond 16bits

4) 16 bit Subtraction: 2345H - 4567H

$$\begin{array}{r} 2345H \\ (-)\ 4567H \\ \hline -2222H \end{array}$$

10000H - 2222H = DDDEH

Result = DDDEH, CY=1

5) 8 bit Multiplication: 14H X 06H

14H = 20

06H = 6

14H X 06H = 14H + 14H + 14H + 14H + 14H + 14H

= 78H

Result = 78H (20 X 6) = 120

b) 8-bit Division 48H ÷ 06H

48H = 72

06H = 6

72 ÷ 6 = 12

12 Decimal = 0CH

=> Quotient(Q) = 0CH

=> Remainder(R) = 00H

c) 16-Bit multiplication : 2345H x 0003H

2345H x 3 = 2345H + 2345H + 2345H

```
  2345H
  2345H
+)2345H
------
  69CFH
```

Result = 000069CFH

d) 16 bit Division: 456FH ÷ 123FH

456FH - 1234 = 3333H

3333H - 1234H = 20FFH

20FFH - 1234H = 0ECBH

0ECBH - 1234H CO

Result:

Quotient (Q) = 0003H

Remainder (R) = 0ECBH

verification:

3 X 1234H + 0ECBH = 456FH.